

Employee Salary Prediction System

1. Introduction

In modern corporate environments, establishing a fair, transparent, and competitive compensation structure is critical for retaining talent and maintaining organizational health. Traditionally, determining employee salaries has often relied on subjective negotiations or fragmented data, which can introduce unconscious bias and pay disparity.

The Employee Salary Prediction System leverages Machine Learning (ML) to automate and standardize salary estimation. By analyzing historical workforce attributes—such as education, job tenure, role requirements, and performance evaluations—the system provides objective, data-driven compensation benchmarks.

2. Objective

The primary goals of this project are:

- **Accurate Salary Estimation (Regression):** Develop a predictive model capable of determining an individual's exact market salary based on specific demographic and professional input parameters.
- **Automated Tendering & Bracketing (Classification):** Categorize compensation packages into explicit tiers (*Low, Medium, High*) to assist Human Resource (HR) departments with budget forecasting and talent tiering.
- **Operational Dashboard Deployment:** Build an accessible, production-ready front-end interface where HR practitioners can easily input employee data to instantly generate salary profiles.

3. Overview of the System

The application is structured around an end-to-end Machine Learning pipeline. The workflow begins with custom data engineering, moves through automated cleaning and algorithmic training, and finishes at a web interface.

The system uses a custom dataset of 250 rows containing 10 key features: `Employee_ID`, `Age`, `Gender`, `Education`, `Department`, `Experience`, `Working_Hours`, `Performance_Score`, `Projects_Completed`, and `Salary`.

The underlying architectural logic connects specific worker qualifications directly to financial scaling rules, ensuring that the machine learning models have clean, logical patterns to learn from.

4. Data Preprocessing

Raw organizational data is rarely ready for machine learning algorithms. The data preprocessing layer ensures that the models receive clean numerical inputs and that different scales do not skew predictions.

The preprocessing pipeline executes five major tasks:

1. **Imputation Check:** Automatically scans for empty entries, filling numeric variables with the column median and categorical attributes with the column mode to prevent program crashes.
2. **Deduplication:** Identifies and drops structurally identical records to avoid artificial overfitting.
3. **Feature Filtering:** Drops the unique tracking column `Employee_ID`, as high-cardinality arbitrary identifiers contain no predictive mathematical value.
4. **Categorical Encoding:** Applies `OneHotEncoder(drop='first')` to translate string features (Gender, Education, Department) into binary flags (\$0\$ or \$1\$) while avoiding the dummy variable trap.
5. **Feature Scaling:** Uses `StandardScaler` to transform numeric data like Age, Experience, and `Working_Hours` into a standardized format with a mean of \$0\$ and a variance of \$1\$. This prevents large raw numbers from disproportionately dominating model weights.
6. **Train-Test Partition:** Splits the treated vectors into an 80% Training Set to fit the learning parameters and a 20% Test Set to measure accuracy on unseen data.

5. Data Visualization

Visual analytics are integrated into the system to explore data relationships and validate model behaviors. Four primary graphs are used to analyze the dataset:

- **Salary Distribution:** A histogram with a Kernel Density Estimate (KDE) line that maps the spread of compensation across the workforce, checking for realistic economic skews.
- **Correlation Heatmap:** A visual matrix showing the linear relationship strength between numerical variables. It verifies that features like Experience and `Projects_Completed` move hand-in-hand with Salary.
- **Experience vs. Salary Scatter Plot:** Plots individual employee markers across a Cartesian graph to show exactly how compensation scales over time, using color-coded groupings to highlight the compounding impact of higher education levels.
- **Department-Wise Comparison:** Uses box plots to visualize salary ranges, median pay, and variance spreads across different business units (e.g., highlighting structurally higher starting baselines in IT and Finance).

6. Machine Learning Algorithms

To fully explore the predictive problem, the backend engine evaluates two distinct categories of machine learning paradigms:

Continuous Value Predictors (Regression)

- **Linear Regression:** Fits a linear hyperplane through the standardized multi-dimensional feature space. It serves as an excellent structural model for this system because our data relies on clean, compounding base-pay multipliers.
- **Random Forest Regressor:** An ensemble approach that constructs a forest of independent decision trees. It makes final exact salary predictions by averaging the continuous numeric outputs of individual leaf nodes, capturing non-linear interactions between variables.

Categorical Tiers (Classification)

- **Logistic Regression:** Applies a multinomial log-odds framework to calculate the probability that an employee belongs to the *Low*, *Medium*, or *High* salary bracket.
- **Random Forest Classifier:** Uses threshold logic boundaries across branching decision criteria to categorize personnel profiles into explicit corporate compensation tiers.

7. Model Evaluation

Model performance is validated systematically across both tasks using standard validation metrics on the testing subset:

Regression Benchmarks

- **Mean Absolute Error (MAE):** Calculates the average absolute dollar amount that predictions miss the real target salary.
- **Root Mean Squared Error (RMSE):** Measures prediction error magnitude, penalizing larger deviations more heavily to identify outlier predictions.
- **R^2 Score (Coefficient of Determination):** Measures the proportion of real-world data variance captured by the model. The models score ≥ 0.94 , proving strong predictive reliability.

Classification Benchmarks

- **Accuracy:** Evaluates the overall percentage of correct salary tier assignments.
- **Confusion Matrix:** Plots True Labels against Predicted Labels to pinpoint exactly where classification errors occur (e.g., misclassifying a *High Salary* executive as *Medium Salary*).

8. Front-End Development

The front-end is built using Streamlit to create an interactive web application that bridges data science with user operations.

The user interface is split into two functional panels:

- **Data Entry Form (Left Panel):** A responsive, single-click submission block featuring dropdown menus, sliding input adjustments, and numerical input fields. This lets users quickly build an input profile mapping directly to the underlying model criteria.
- **Analytical Dashboard (Right Panel):** A tabbed viewing workspace that displays data frames, loads interactive charts dynamically via Seaborn, and displays real-time performance tables comparing model accuracies.

9. Conclusion

The **Employee Salary Prediction System** successfully demonstrates how modern data pipelines and machine learning techniques can be applied to corporate Human Resources operations. By transitioning from a legacy, intuition-based compensation structure to an algorithmic framework, organizations can minimize systemic bias, optimize departmental budget allocation, and ensure fair pay equity across all demographics.

Through systematic evaluation, the system highlights several key engineering takeaways:

- **Algorithmic Synergy:** The **Linear Regression** model excelled at continuous salary estimation because the underlying dataset follows systematic, compounding corporate rules. Conversely, the tree-based **Random Forest Classifier** proved optimal for tier classification by isolating non-linear interaction thresholds (such as the combined impact of holding a PhD and having high performance marks).
- **Pipeline Integrity:** The project emphasizes that model accuracy relies heavily on proper data preprocessing. Standardizing numerical scales and applying robust encoding rules are critical steps for preventing model bias.
- **Operational Value:** Deploying the models via a **Streamlit front-end** bridges the gap between complex data science and everyday business tools. This ensures that non-technical HR managers can leverage predictive analytics through an intuitive dashboard to make data-driven hiring and promotion decisions.
- Ultimately, this architecture serves as a scalable foundation. In production environments, it can be expanded by integrating live ERP systems, tracking market inflation variables, and incorporating more granular skills data to further refine organizational compensation strategies.